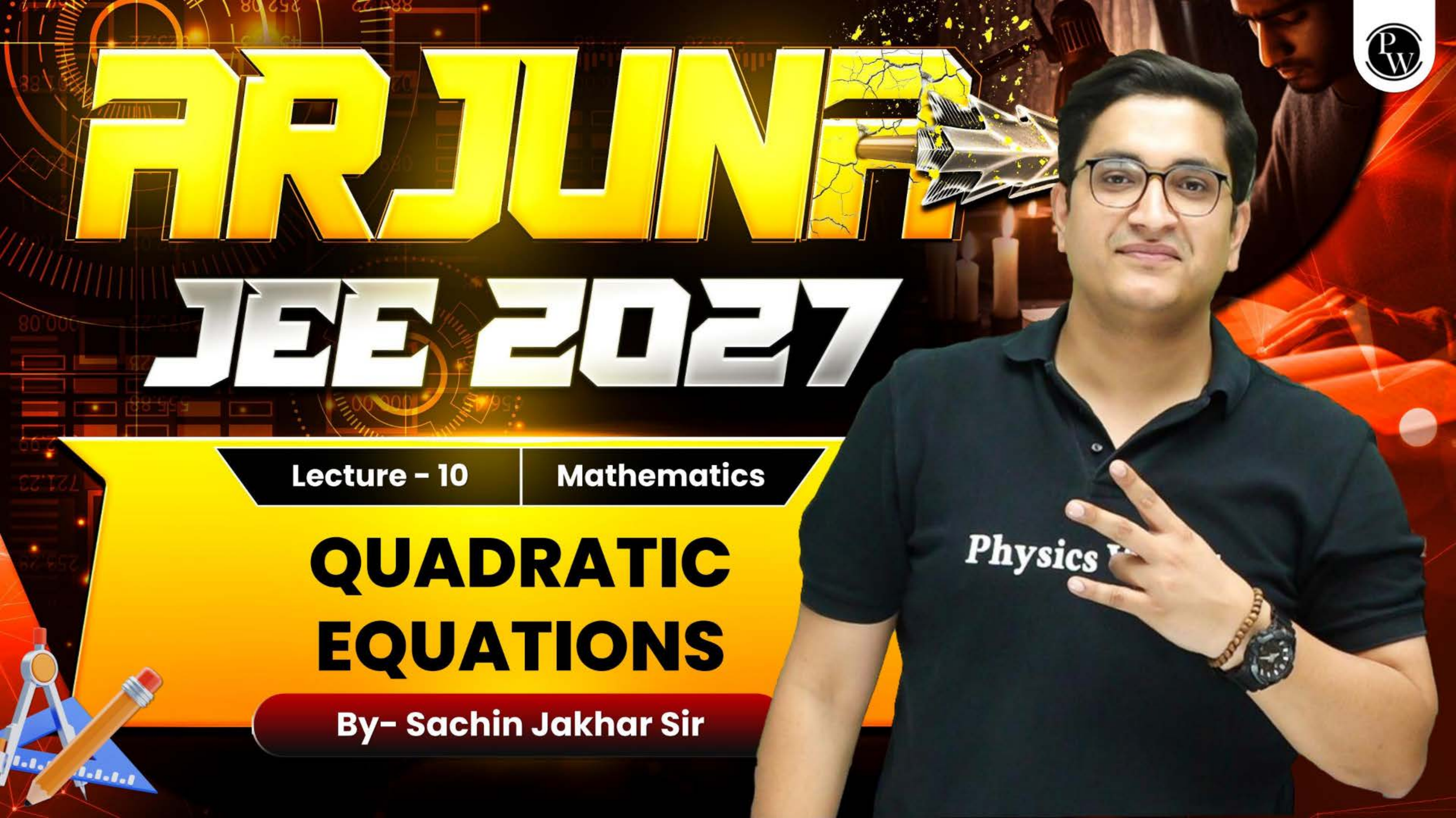
ARJUNA JEE 2027

Lecture - 10 | Mathematics

QUADRATIC EQUATIONS

By- Sachin Jakhar Sir



Topics *to be covered*

A Range of Rational Functions

B Common Root Condition

C Question Practice

D





LAST CLASS REVISION

Answer the following Questions:

Quadratic: $y = ax^2 + bx + c$

(i) If $a > 0 \Rightarrow$ $\text{Range}(y) \in \left[\frac{-D}{4a}, \infty\right)$

(ii) If $a < 0 \Rightarrow$ $\text{Range}(y) \in \left(-\infty, \frac{-D}{4a}\right]$

if $x \in R$

$x \in [p, q]$

Quad. Range?

Graphical Method

Question Practice

Challenger Question [IIT-JEE]

If $a < b < c$ and $a, b, c \in R$ then comment on nature of roots of following:

(i) $f(x) = (x-a)(x-b) - 1$

(ii) $f(x) = (x-a)(x-b) + (x-b)(x-c) + (x-c)(x-a)$ $= 3x^2 + \ldots\ldots$

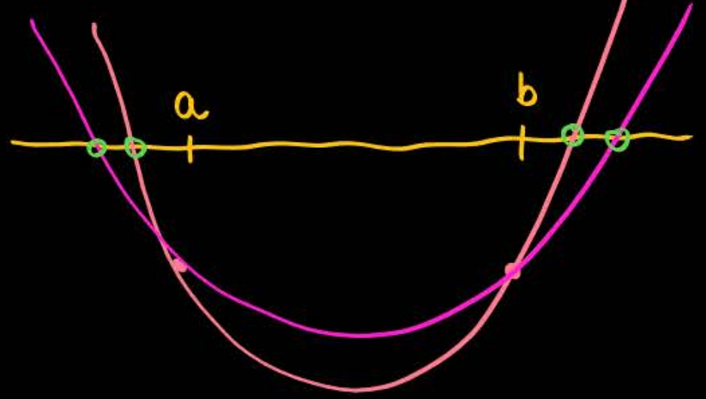
Graph :- $f(x) = 1x^2 - (a+b)x + ab - 1$

$f(a) = -1, \quad f(b) = -1.$

2 R & D Root.

1 Root $> b$

1 Root $< a$.



$x = a$

$f(a) = (a-b)(a-c) \rightarrow +$

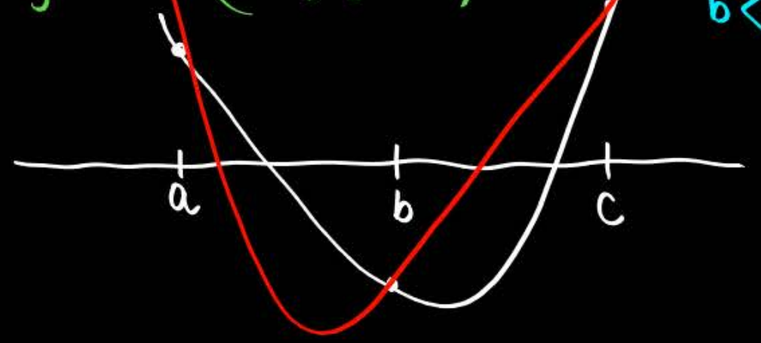
$f(b) = (b-c)(b-a) \rightarrow -$

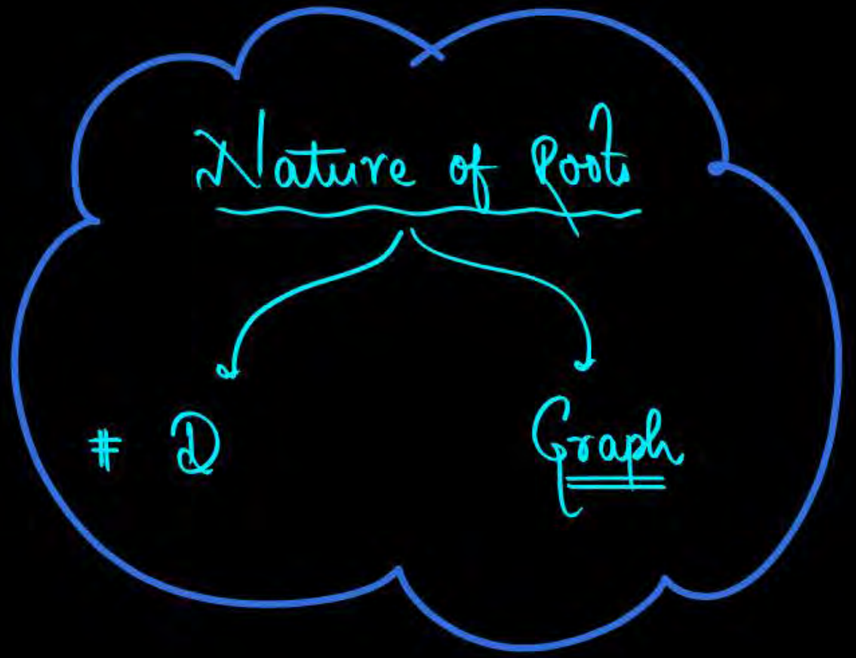
$f(c) = (c-a)(c-b) \rightarrow +$

2 R & D Root.

$a <$ 1 Root $< b$

$b <$ 1 Root $< c$




Nature of Posts
D
Graph

Question

Find range for following functions :

(i) $f(x) = -x^2 + 4x - 6$

(ii) $f(x) = -x^2 + 4x - 6, x \in [-3, 1)$

(iii) $f(x) = -x^2 + 4x - 6, x \in (0, 4]$

Ans. : (i) $y \in (-\infty, -2]$, (ii) $y \in [-27, -3)$, (iii) $y \in [-6, -2]$

Question

Find range for following functions :

DIBY - (A)

(i) $f(x) = 2x^2 - 3x + 5, x \in [-1, 1]$ ✓

(ii) $f(x) = x^2 - x - 8, x \in (-\infty, -2] \cup [2, \infty)$ ✓

(iii) $f(x) = 5 \cdot 4^x - 2^{x+3} + 1$

$f(x) = 5(2^x)^2 - 2^x \cdot 2^3 + 1$

Say: $2^x = t$

+ve

$2^x = t$

$t \in (0, \infty)$

Quad. in Terms of any other func?

$f(t) = 5t^2 - 8t + 1$, $t \in (0, \infty)$

Ans. : (i) $y \in \left[\frac{31}{8}, 10\right]$, (ii) $y \in [-6, \infty)$, (iii) $y \in \left[-\frac{11}{5}, \infty\right)$

$f(\theta) = 2\sin^2\theta - 6\sin\theta + 7$, Range ??

$[-1,1]$

$\sin\theta = t$

$$f(t) = 2t^2 - 6t + 7 \,, \quad t \in [-1,1]$$

DIBY — (B)

$$y = \frac{ax+b}{cx+d}$$

$x \in R - \left\{-\frac{d}{c}\right\}$

$y \in R - \left\{\frac{a}{c}\right\}$

ex:- $y = \frac{2x+7}{5-3x}$

$x \in R - \left\{\frac{5}{3}\right\}$

$y \in R - \left\{-\frac{2}{3}\right\}$

ex:- $y = \frac{x-7}{14-2x}$

$x \in R - \{7\}$

$y \in R - \left\{-\frac{1}{2}\right\}$ ×

$y = \frac{x-7}{-2(x-7)} = -\frac{1}{2}$

$y \in \left\{-\frac{1}{2}\right\}$ ✓

Note :-

$f(m) \cdot f(n) < 0$

- 1 Root lies b/w m & n
- 2 Real & distinct roots.



RANGE OF RATIONAL FUNCTION

Format : $y = \frac{\text{Quad.}}{\text{Quad.}}$ *or* $\frac{\text{Linear}}{\text{Quad.}}$ *or* $\frac{\text{Quad.}}{\text{Linear}}$ *or* $\frac{\text{Constant}}{\text{Quad.}}$

Step-I : Cross-multiply & form a Quadratic in 'x',

Step-II : Make $D \geq 0$ or above quadratic. (as $x \in$ Real)

Step-III : Solve above inequality using Wavy Curve.(This answer is not final)

Step-IV: **Check your Answer**

Make coefficient of $x^2 = 0$

Obtain 'y' & put that value of 'y' in equation (1)

If you get any 'x' $\rightarrow$ Then S-III final ans.

If you don't get any 'x' $\rightarrow$ Then remove that 'y' from S-III.

Bumper Question

Find range of following function :

(i) $f(x) = \dfrac{(x+2)(x-1)}{x(x+1)}$

(ii) $f(x) = \dfrac{x^2+x+1}{x^2-x+1} = \dfrac{Q_1}{Q_2}$

(iii) $f(x) = \dfrac{x^2+2x-3}{x^2-8x+7}$

(iv) $f(x) = \dfrac{x^2+1}{x}$

(2) $y = \dfrac{x^2+x+1}{x^2-x+1} \in \left[\frac{1}{3}, 3\right]$

$x^2y - xy + y = x^2 + x + 1$

Real

$(y-1)x^2 - (y+1)x + (y-1) = 0$ Quad. in x.

$\Rightarrow D \geq 0$

$(-(y+1))^2 - 4(y-1)(y-1) \geq 0$

$y^2 + 2y + 1 - 4(y^2 - 2y + 1) \geq 0$

$-3y^2 + 10y - 3 \geq 0$

$3y^2 - 10y + 3 \leq 0$

$3y^2 - 9y - y + 3 \leq 0$

$(3y-1)(y-3) \leq 0$

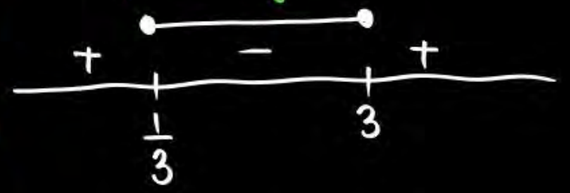
$$y = \frac{x^2+x+1}{x^2-x+1} \quad \text{------ (1)}$$

$$(y-1)x^2 - (y+1)x + (y-1) = 0$$

$y \neq 1$

$\Rightarrow D \geq 0$

$$(3y-1)(y-3) \leq 0$$



$$\# \; y \in \left[\frac{1}{3}, 3\right]$$ fnf.

Check:

Coeff. of $x^2 = y - 1 = 0$

$y = 1$

$$y = \frac{x^2+x+1}{x^2-x+1}$$

$$1 = \frac{x^2+x+1}{x^2-x+1}$$

$$x^2 - x + 1 = x^2 + x + 1$$

$$0 = 2x$$

$0 = x$ ✓

Bumper Question

Find range of following function :

(i) $f(x) = \dfrac{(x+2)(x-1)}{x(x+1)}$

(ii) ~~(ii)~~ $f(x) = \dfrac{x^2+x+1}{x^2-x+1}$

(iii) $f(x) = \dfrac{x^2+2x-3}{x^2-8x+7}$

(iv) $f(x) = \dfrac{x^2+1}{x}$

(i) $y = \dfrac{x^2+x-2}{x^2+x}$ — (1)

$x^2y + xy = x^2 + x - 2$

Quad. $(y-1)x^2 + (y-1)x + 2 = 0$

$D \geq 0$

$(y-1)^2 - 4(y-1) \times 2 \geq 0$

$(y-1)\{y-1-8\} \geq 0$

$(y-1)(y-9) \geq 0$



$y \in (-\infty, 1] \cup [9, \infty)$

Check :- $y - 1 = 0 \Rightarrow y = 1$ → Remove.

$1 = \dfrac{x^2+x-2}{x^2+x}$

$x^2 + x = x^2 + x - 2$

$0 = -2$ → No value of x obtained

Ans : $y \in (-\infty, 1) \cup [9, \infty)$

Bumper Question

Find range of following function :

(i) $f(x) = \dfrac{(x+2)(x-1)}{x(x+1)}$

(ii) ~~(ii)~~ $f(x) = \dfrac{x^2+x+1}{x^2-x+1}$

(iii) $f(x) = \dfrac{x^2+2x-3}{x^2-8x+7}$

Impost.

$= \dfrac{(x+3)(x-1)}{(x-1)(x-7)}$

$x = 1$

$x - 1 = 0$

$\# \; x \in R - \{1, 7\}$

$y = \dfrac{x+3}{x-7}$

value of y corresp. to cancelled factor

$y \in R - \left\{1, -\dfrac{2}{3}\right\}$

(iv) $f(x) = \dfrac{x^2+1}{x}$

ex:-

$$y = \frac{(2x-1)(x+3)}{(x-3)(2x-1)}$$

$x \in R - \left\{3, \frac{1}{2}\right\}$

$$y = \frac{x+3}{x-3}$$

$x = \frac{1}{2} \rightarrow y = \frac{-7}{5}$

$x = \frac{1}{2}$

$y \in R - \left\{1, \frac{-7}{5}\right\}$

(iv) $y = \frac{x^2+1}{x}$ — Quad. / Linear.

$xy = x^2 + 1$

$1x^2 - yx + 1 = 0$

$D \geq 0$

$(-y)^2 - 4 \times 1 \geq 0$

$y^2 - 4 \geq 0$

$(y-2)(y+2) \geq 0$



$y \in (-\infty, -2] \cup [2, \infty)$

Question [JEE (Adv.)-2021]

Let $f: \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \dfrac{x^2 - 3x - 6}{x^2 + 2x + 4}$. Find range.

$$y = \frac{x^2 - 3x - 6}{x^2 + 2x + 4}$$

$$x^2y + 2xy + 4y = x^2 - 3x - 6$$

$$(y-1)x^2 + (2y+3)x + (4y+6) = 0$$

$D \geq 0$

$$(2y+3)^2 - 4(y-1)(4y+6) \geq 0$$

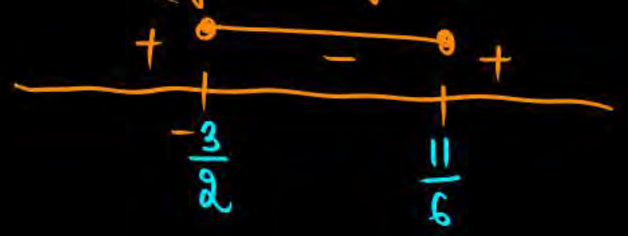
$$-12y^2 + 4y + 33 \geq 0$$

$$12y^2 - 4y - 33 \leq 0$$

$$12y^2 + 18y - 22y - 33 \leq 0$$

$$6y(2y+3) - 11(2y+3) \leq 0$$

$$(2y+3)(6y-11) \leq 0$$



$$y = \frac{x^2 - 3x - 6}{x^2 + 2x + 4}$$

$$(y-1)x^2 + (2y+3)x + (4y+6) = 0$$

$$D \geq 0$$

$$(2y+3)(6y-11) \leq 0$$



$$y \in \left[\frac{-3}{2}, \frac{11}{2}\right] \text{ fnf.}$$

• Check:

$$y - 1 = 0$$

$$y = 1$$

$$1 = \frac{x^2 - 3x - 6}{x^2 + 2x + 4}$$

$$x^2 + 2x + 4 = x^2 - 3x - 6$$

$$5x = -10$$

$$x = -2 \checkmark$$

Critical Thinking Question (CTQ)

If the range of the function $f(x) = \frac{x^2 + ax + b}{x^2 + 2x + 3}$ is $[-5, 4]$, $a, b \in N$, then find the value of $(a^2 + b^2)$



S^4

- $y = ax^2+bx+c$
  - Graph
  - Range. → $\left[\frac{-D}{4a}, \infty\right)$ or $\left(-\infty, \frac{-D}{4a}\right]$ } $x \in R$

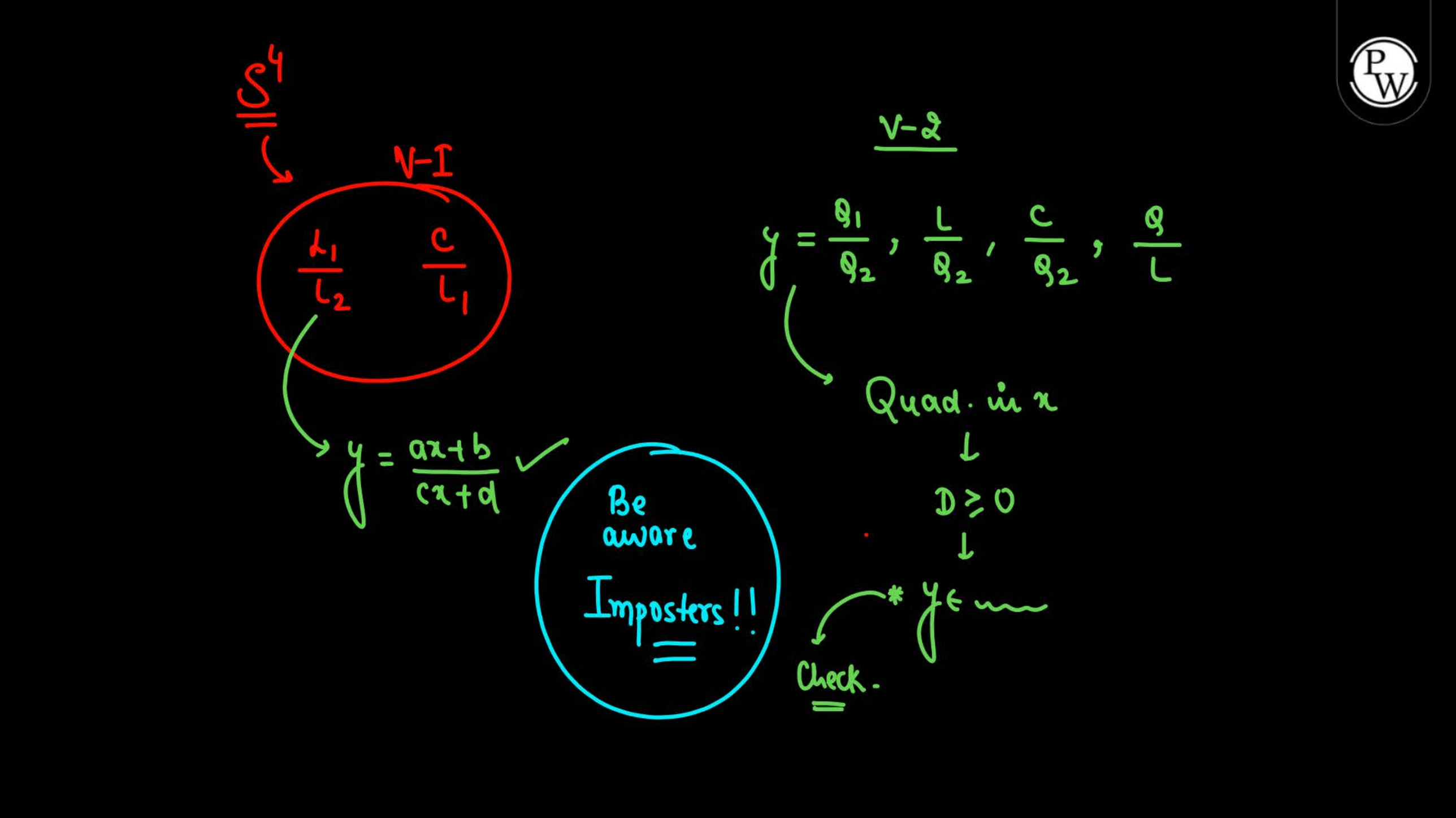
- $y = ax^2+bx+c$, $x \in [p, q)$
  - Graphical.

- Range?
- $y = 2(h(x))^2 + 3h(x) - 7$
  - Quad. in $h(x)$
  - $h(x) = t$

$y = 2t^2+3t-7$, $t \in$ ...

↓

Range.


ICSI
V-I
$\frac{a_1}{L_2}$
$\frac{c}{L_1}$
$y = \frac{ax+b}{cx+d}$
V-2
$y = \frac{a_1}{a_2}, \frac{L}{a_2}, \frac{c}{a_2}, \frac{a}{L}$
Quad. in x
$D \geq 0$
* y ∈ ...
Check.
Be aware
Imposters!!

COMMON ROOT CONDITION

Case – I : If both roots are common

$a_1x^2 + b_1x + c_1 = 0$ — α, β

$a_2x^2 + b_2x + c_2 = 0$ — α, β

ex:- $x^2 - 5x + 6 = 0$ — 2, 3

$7x^2 - 35x + 42 = 0$ — 2, 3

(#) Condⁿ :- $\boxed{\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}}$

Ques:

$$\left.\begin{aligned} ax^2+bx+c=0 \\ 7x^2-6x+5=0 \end{aligned}\right\}$$ both Roots are common

$$\frac{a}{7}=\frac{b}{-6}=\frac{c}{5}=k$$

$a=7k$

$b=-6k$

$c=5k$

Then $\dfrac{a^3+b^3+c^3}{3abc}=??$

$$\frac{7^3k^3-6^3k^3+5^3k^3}{3\times(7k)(-6k)(5k)}=\left(\frac{7^3-6^3+5^3}{-630}\right)$$

Proof:

$a_1x^2 + b_1x + c_1 = 0$ — roots α, β

$a_2x^2 + b_2x + c_2 = 0$ — roots α, γ

$a_1\alpha^2 + b_1\alpha + c_1 = 0 \xrightarrow{\times a_2} \underline{a_2a_1\alpha^2} + a_2b_1\alpha + a_2c_1 = 0$

$a_2\alpha^2 + b_2\alpha + c_2 = 0 \xrightarrow{\times a_1} \underline{a_1a_2\alpha^2} + a_1b_2\alpha + a_1c_2 = 0$

$\ominus$

$\alpha(a_2b_1 - a_1b_2) + (a_2c_1 - a_1c_2) = 0$

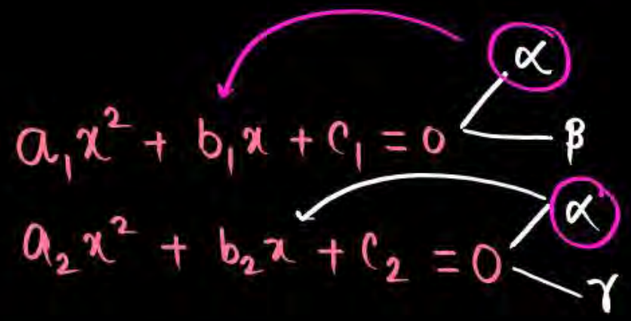
$$\alpha = \frac{a_1c_2 - a_2c_1}{(a_2b_1 - a_1b_2)}$$

$\times b_2$: $b_2a_1\alpha^2 + \boxed{b_2b_1\alpha} + b_2c_1 = 0$

$\times b_1$: $a_2b_1\alpha^2 + \boxed{b_2b_1\alpha} + b_1c_2 = 0$

$\ominus$

$$\alpha^2 = \frac{b_1c_2 - b_2c_1}{(b_2a_1 - a_2b_1)}$$

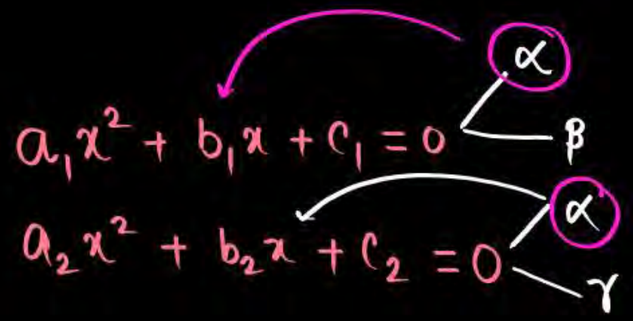


$$\alpha = \frac{a_1c_2 - a_2c_1}{(a_2b_1 - a_1b_2)}$$

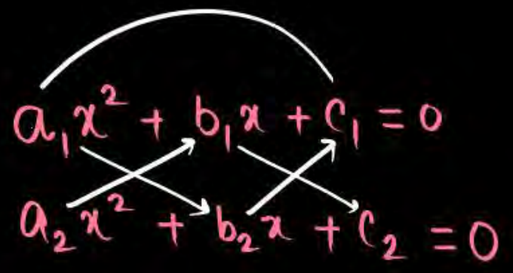
$$\alpha^2 = \frac{b_1c_2 - b_2c_1}{(b_2a_1 - a_2b_1)}$$

$$\left(\frac{a_1c_2 - a_2c_1}{(a_2b_1 - a_1b_2)}\right)^2 = \frac{b_1c_2 - b_2c_1}{(b_2a_1 - a_2b_1)}$$

$$\frac{(a_1c_2 - a_2c_1)^2}{(a_1b_2 - a_2b_1)^2} = \frac{b_1c_2 - b_2c_1}{(b_2a_1 - a_2b_1)} \Rightarrow (a_1c_2 - a_2c_1)^2 = (b_1c_2 - b_2c_1)(a_1b_2 - a_2b_1)$$



$$\boxed{(a_1c_2 - a_2c_1)^2 = (b_1c_2 - b_2c_1)(a_1b_2 - a_2b_1)}$$



$$(a_1b_2 - a_2b_1)\ (b_1c_2 - b_2c_1) = (a_1c_2 - a_2c_1)^2$$

$$a_1x^2 + b_1x + c_1 = 0$$

$$a_2x^2 + b_2x + c_2 = 0$$

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \cdot \begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix} = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}^2$$

$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

$\begin{vmatrix} 2 & 4 \\ 5 & 3 \end{vmatrix} = 6 - 20 = -14.$

$\begin{vmatrix} 2 & -1 \\ 4 & -3 \end{vmatrix} = -6 - (-4) = -2.$

$\begin{vmatrix} 2 & -3 \\ 8 & 5 \end{vmatrix} = 10 - (-24)$

$= 34.$

COMMON ROOT CONDITION

Case – II : If atleast one root are common

$$a_1x^2 + b_1x + c_1 = 0$$

$$a_2x^2 + b_2x + c_2 = 0$$

$$(a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1) = (a_1c_2 - a_2c_1)^2$$

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \cdot \begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix} = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}^2$$

Atleast 1 Root common

1 Root common

2 Roots common.

Question

Find the value of p if the equation $3x^2 - 2x + p = 0$ and $6x^2 - 17x + 12 = 0$ have a common roots.

$$3x^2 - 2x + p = 0$$
$$6x^2 - 17x + 12 = 0$$

$$\begin{vmatrix} 3 & -2 \\ 6 & -17 \end{vmatrix} \times \begin{vmatrix} -2 & p \\ -17 & 12 \end{vmatrix} = \begin{vmatrix} 3 & p \\ 6 & 12 \end{vmatrix}^2$$

$$(-51+12)(-24+17p) = (36-6p)^2$$

$$\# \; p = \checkmark$$

Question [JEE Mains-2023]

If the value of real number $a > 0$ for which $x^2 - 5ax + 1 = 0$ and $x^2 - ax - 5 = 0$ have a common real roots is $\frac{3}{\sqrt{2\beta}}$, then β is equal to



DO IT BY YOURSELF (DIBY)

GRAPH OF QUADRATIC

DIBY-36

Consider the graph $f(x) = ax^2 + bx + c$ and comment on sign of following parts:

(i) (abc) is _____.

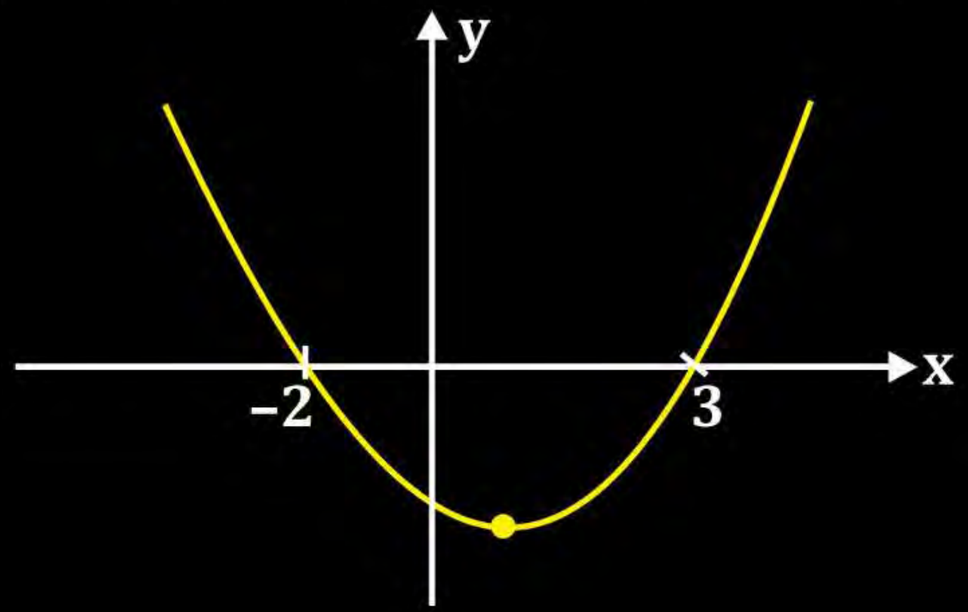
(ii) $(4a + 2b + c)$ is _____.

(iii) for all $x > 3, f(x)$ is _____.

(iv) For all $-2 < x < 3, f(x)$ is _____.

(v) D is _____.

(vi) $(a - b + c)\cdot(9a - 3b + c)$ is _____.



Ans. :

(i) +ve, (ii) –ve, (iii) +ve, (iv) –ve, (v) +ve, (vi) –ve

GRAPH OF QUADRATIC

DIBY-37(Challenger)

Draw the graph of quadratic $(y = ax^2 + bc + c)$ in each part with given conditions:

(i) Roots are 3 & 7 and crosses y-axis at (0, −2)

(ii) Imaginary roots with vertex at $\left(-\frac{3}{4}, \frac{1}{2}\right)$ and $a - b + c = 2$.

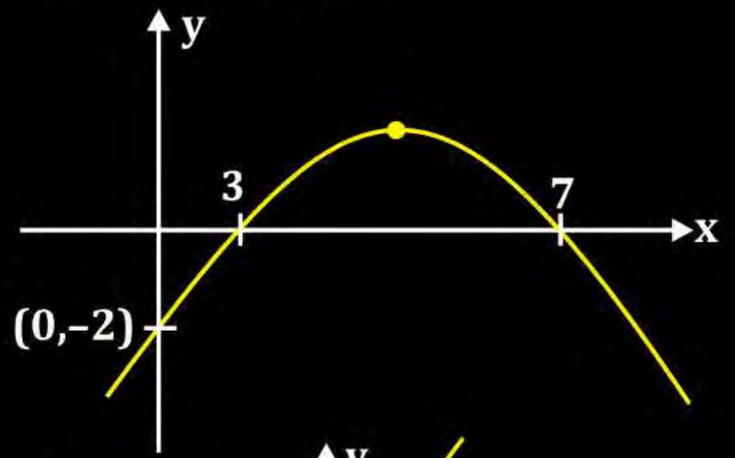
(iii) Touches x-axis at $x = -3$ and value at $x = 2$ is −8.

Also find value of a, b & c in each case.

GRAPH OF QUADRATIC

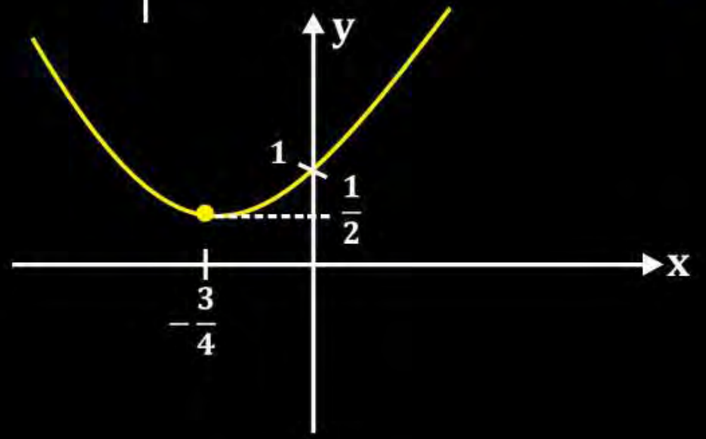
DIBY-37(Answer)

(i)



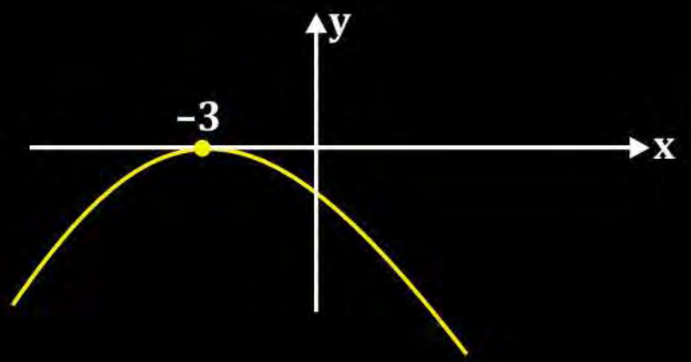
$a = -\frac{2}{21}, b = \frac{20}{21}, c = -2$

(ii)



$a = 24, b = 36, c = 14$

(iii)



$a = \frac{-8}{25}, b = \frac{-48}{25}, c = \frac{-72}{25}$

DIBY-38

If the equation $ax^2 + bc + c = 0, (a > 0)$ has two real roots α and β such that $\alpha < -2$ and $\beta > 2$, then which of the following statements is/are true?

(A) $a - |b| + c < 0$

(B) $c < 0,\ b^2 - 4ac > 0$

(C) $4a - 2|b| + c < 0$

(D) $9a - 3|b| + c < 0$

[Ans. : A, B, C]

DIBY-39

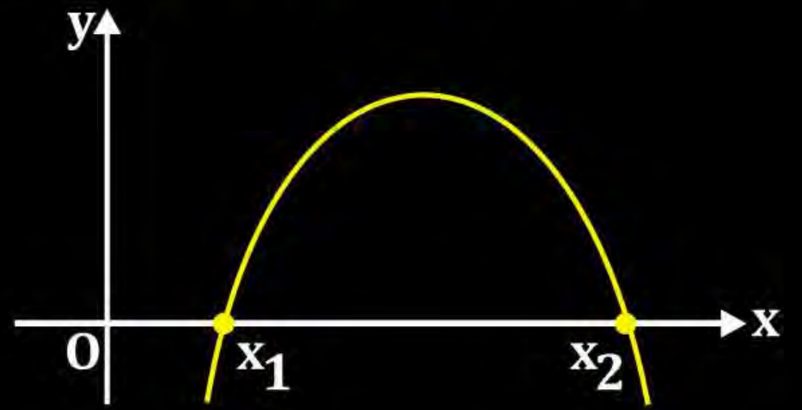
The adjoining figure shows the graph of $y = ax^2 + bx + c$. Then

(A) $a < 0$

(B) $b^2 < 4ax$

(C) $c > 0$

(D) a and b are of opposite sign



[Ans. : A, D]

RANGE OF QUADRATIC

DIBY-40

Find the range of $f(x)$:

(i) $f(x) = 2x^2 - 3x + 2$

(ii) $f(x) = 2x^2 - 3x + 2, x \in [0, 2]$

[Ans. : (i) $\left[\frac{7}{8}, \infty\right)$, (ii) $\left[\frac{7}{8}, 4\right]$]

DIBY-41

Find range of following functions:

(i) $f(x) = 3x^2 - 2x - 7$

(ii) $f(x) = 3x^2 - 2x - 7, x \in (0, 5]$

(iii) $f(x) = 3x^2 - 2x - 7, x \in [-6, -1]$

[Ans. : (i) $\left[-\frac{22}{3}, \infty\right)$, (ii) $[-22/3, 58]$, (iii) $[-2, 113]$]

RANGE OF RATIONAL FUNCTION

DIBY-42

If the range of the function $f(x) = \frac{5-x}{x^2-3x+2}, x \neq 1, 2$, is $(-\infty, \alpha] \cup [\beta, \infty)$, then $\alpha^2 + \beta^2$ is equal to:

[JEE Mains-2025]

(A) 190 (B) 192 (C) 188 (D) 194

[Ans. : (D)]

DIBY-43

Let $f: \mathbb{R} - \{2, 6\} \to \mathbb{R}$ be real valued function defined as $f(x) = \frac{x^2+2x+1}{x^2-8x+12}$. Then range of f is

[JEE Mains-2023]

(A) $\left(-\infty, -\frac{21}{4}\right] \cup [0, \infty)$

(B) $\left(-\infty, -\frac{21}{4}\right) \cup (0, \infty)$

(c) $\left(-\infty, -\frac{21}{4}\right] \cup \left[\frac{21}{4}, \infty\right)$

(D) $\left(-\infty, -\frac{21}{4}\right] \cup [1, \infty)$

[Ans. : (A)]

RANGE OF RATIONAL FUNCTION

DIBY-44

Let the sum of the maximum and the minimum values of the function $f(x) = \frac{2x^2-3x+8}{2x^2+3x+8}$ be $\frac{m}{n}$, where $\gcd(m,n) = 1$. Then $m+n$ is equal to

[JEE Mains-2024]

(A) 195 (B) 201 (C) 217 (D) 182

[Ans. : (B)]

DIBY-45

Show that if x is real then $\frac{x^2-3x+4}{x^2+3x+4}$ lies from 1/7 and 7.

DIBY-46

Find the range of : $f(x) = \frac{x^2+x+2}{x^2+x+1}, x \in R$ [IITJEE-2003]

[Ans. : $\left(1, \frac{7}{3}\right]$]

RANGE OF RATIONAL FUNCTION

DIBY-47

We call 'p' a good number if the inequality $\frac{2x^2+2x+3}{x^2+x+1} \leq p$ is satisfied for any real x. Find the smallest integral good number.

[Ans. : 4]

DIBY-48

If α, β are the roots of equation $x^2 + x - 2 = 0$, then find the value of

$$\frac{\alpha\beta^4(\beta+1)^4+\beta\alpha^4(\alpha+1)^4}{\alpha^2+\beta^2+\alpha+\beta}.$$

DIBY-49

The minimum value of the sum of the squares of the roots of $x^2 + (3 - a)x + 1 = 2a$ is:

[Ans. : 6]

DIBY-50

The number of real solutions of the equation $3\left(x^2+\frac{1}{x^2}\right)-2\left(x+\frac{1}{x}\right)+5=0$, is

[JEE Mains-2023]

(A) 4 (B) 0 (C) 3 (D) 2

Ans. : (B)

COMMON ROOT CONDITION

DIBY-51

If $x^2+4ax+3=0$ and $2x^2+3ax-9=0$ have a common root then the value of $|a|$ is

(A) 1 (B) 0 (C) -1 (D) None of these

[Ans. : A]



Homework

Re-attempt all Class Examples

DPP & DIBY

Module: (3 sittings of 1 HR each)

Exercise-02(Prabal): 1 to 10,13,16,18 to 25.

Exercise-03(Setu): 1 to 30 (except-13,16,17,21,25,26,29)

It is never about end result

It's always about Journey

Thank You

#futureIITians